

Ravi Mosalpuri

✉ ravinmosalpuri@gmail.com | 📞 +91-8112288643 | 🔗 [linkedin.com/in/ravi-mosalpuri/](https://www.linkedin.com/in/ravi-mosalpuri/) | 🐙 github.com/ravimosalpuri

About Me

I'm a skilled software engineer with over 5 years of experience. I'm passionate about solving complex problems through code and am committed to delivering high-quality solutions. With a strong attention to detail, I thrive in fast-paced environments and continuously seek opportunities to learn.

Skills

Languages: C++ (Modern, STL), C#, Python, SQL, PostgreSQL

Technologies & Tools: CI/CD, Perforce, Git, Jira, Confluence, TeamCity, GitLab, GitHub Actions, SonarQube, NUnit

Libraries & APIs: Pandas, Numpy, scikit-learn, TensorFlow, PyTorch, FastAPI, OpenMP, MPI

Soft Skills: Communication, Teamwork, Problem-solving, Leadership

Work Experience

Barclays - AVP (C++ Developer)

Jan 2026 - Present

- Joined the Counterparty Credit Risk team as a C++ developer, enhancing reliability and performance in risk evaluation.
- Developed and optimized quantitative models for equity and derivatives, enhancing performance.
- Integrated large language models (LLMs) with C++ libraries, enabling agent-based automation and task execution.
- Developed refined UI for an AI-powered chatbot agent, improving usability and user interaction with backend systems

NatWest Group - Software Engineer

Feb 2024 - Aug 2024

- Contributed to 'Single Trade Representation' project, focusing on .NET-based implementation.
- Delivered a Proof of Concept for Intraday P&L and risk measures, resulting projected savings of \$5M.
- Received individual recognition twice for technical excellence and delivering under tight deadlines.

NatWest Group - Quantitative Developer (C++ Developer)

Jan 2023 - Jan 2024

- Developed backend C++ libraries in Front Office team, focusing on performance and accuracy.
- Spearheaded the integration of several trade-related functionalities within the backend library.
- Implemented unit tests for financial algorithms using Google Test, adding reliability & test coverage.
- Automated Excel data processing workflows by scripting Python tools, reducing manual effort.

Ubisoft - Programmer

July 2019 - Oct 2022

- Gained hands-on experience in development with C++, C#, and Python, applying them across gameplay systems, UI architecture, platform integration, and automation workflows.
- Worked on multiple titles including Prince of Persia: The Sands of Time Remake (AAA title), Monopoly Madness (Turn-based), Roller Champions (Sports), and an unannounced MMORPG.
- Developed in-game UI components, including HUD elements, dynamic menus, and inventory systems using the MVVM architecture to ensure a scalable, modular and maintainable codebase.
- Implemented cross-progression, save systems, and multiplayer features across platforms using C#, REST APIs, and deep linking, enabling seamless player data synchronisation.

Education

University of Exeter, United Kingdom

Sept 2024 - Sept 2025

M.Sc. in Advanced Computer Science - *Distinction with Dean's Commendation award*

Coursework: Machine Learning, High Performance Computing, Data Governance and Ethics

Indian Institute of Technology Bombay, India

July 2015 - July 2019

Bachelor of Technology

Projects

Classification of Histopathology images Using Deep Learning

Feb 2024 - August 2024

- Developed a custom Vision Transformer (ViT-16) in Python for multi-class histopathology image classification.
- Built a robust pipeline with advanced preprocessing, augmentation, and cross-validation across six cancer datasets to maximise generalisability.
- Designed a multi-layer classification head with normalisation and dropout, improving resistance to overfitting and feature transferability.
- Achieved state-of-the-art results on three datasets: 100% accuracy on lung and colon, 99.85% on ALL leukemia.
- Received Best MSc Project Award and Published findings in a research paper ([link](#)).